

University of Hertfordshire

7WCM2034 Fundamentals of Software Engineering Coursework

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Question 1

In modern society, software have become first place, whether it is software that controls traffic lights or those that power a pacemaker, the need for software to run efficiently is vital to our well-being. Software engineering then becomes essential because it enables the design and development of reliable and scalable systems that serves our everyday needs.

Question 2

- a) **Systems Integrator** is accountable for the implementation and integration of software or systems.
- b) SLAs, OLAs, and KPIs requirements are defined by Service Manager.
- c) Quality policy established and deployed by Quality Assurance Manager.
- d) Digital Educator perform effective training in the classroom, online or informally.
- e) Risks, threats, and consequences are evaluated by Cyber Security Manager.
- f) Information Security Manager define and implement procedures linked to IS security.
- g) Staff coordinator and manager is ICT Operations Manager.
- h) Test Specialist develop tools to increase test effectiveness.

Question 3

- a) According to the Software Engineering Body of Knowledge, **Software requirements** expresses the needs and constraints placed on a software product that contribute to the solution of some real world problem.
- b) **Functional requirements** describe the functions that the software is to execute; for example, formatting some text or modulating a signal. They are sometimes known as capabilities or features. A functional requirement can also be described as one for which a finite set of test steps can be written to validate its behavior. **Nonfunctional requirements** are the ones that act to constrain the solution. Nonfunctional requirements are sometimes known as constraints or quality requirements.

- c) **Software design principles** are key notions that provide the basis for many different software design approaches and concepts. Software design principles include abstraction; coupling and cohesion; decomposition and modularization; encapsulation/ information hiding; separation of interface and implementation; sufficiency, completeness, and primitiveness; and separation of concerns.
- d) **Software Maintenance** is defined as the totality of activities required to provide cost-effective support to software. Activities are performed during the predelivery stage as well as during the postdelivery stage.

Question 4

The “Big Picture” of software development can be understood through the Software Development Life Cycle (SDLC), which outlines the key phases involved in building software, and these phases can be mapped to the European ICT Professional Profiles through Software Engineering sub-disciplines.

a) Requirement Gathering and Analysis:

European ICT Professional Profiles:

- Business Analysis
- Solution Architecture
- Systems Analysis

Software Engineering Sub-disciplines:

- Requirements Engineering
- Software Requirements Specification
- Domain Analysis

b) Design:

European ICT Professional Profiles:

- Software Design
- Infrastructure Architecture
- User Experience Design

Software Engineering Sub-disciplines:

- Architectural Design
- Detailed Design
- User Interface Design

c) Implementation:

European ICT Professional Profiles:

- Software Development
- Database Development
- Network Engineering

Software Engineering Sub-disciplines:

- Coding
- Database Implementation
- Configuration Management

d) Testing:

European ICT Professional Profiles:

- Quality Assurance
- Test Engineering
- Security Testing

Software Engineering Sub-disciplines:

- Unit Testing
- Integration Testing
- System Testing

e) Deployment and Maintenance:

European ICT Professional Profiles:

- Service Management
- Technical Support
- IT Operations

Software Engineering Sub-disciplines:

- Deployment Planning
- Release Management

Question 5

SYSTEMS ARCHITECT

A Systems Architect is an ICT professional responsible for defining the overall system structure and ensuring that technical solutions align with business needs. According to ICT role profiles, a role represents a set of responsibilities, competences, and deliverables within the ICT process .

In the **Requirements Stage**, the Systems Architect collaborates with analysts to interpret business requirements and assess feasibility.

In the **Design Phase**, they lead the development of system architecture, data models, and integration structures.

During implementation, they guide developers and ensure the system is built according to the defined architecture.

In the **Testing Phase**, they verify that the system meets quality and performance expectations.

In the **Deployment Phase**, they support system configuration and release. Finally, in **Maintenance Phase**, they oversee system evolution, ensuring continuous improvement and adaptability. - [European ICT professionals role profiles - Part 2: User guides \(2018\) SWEBOK® Guide V3.0](#)

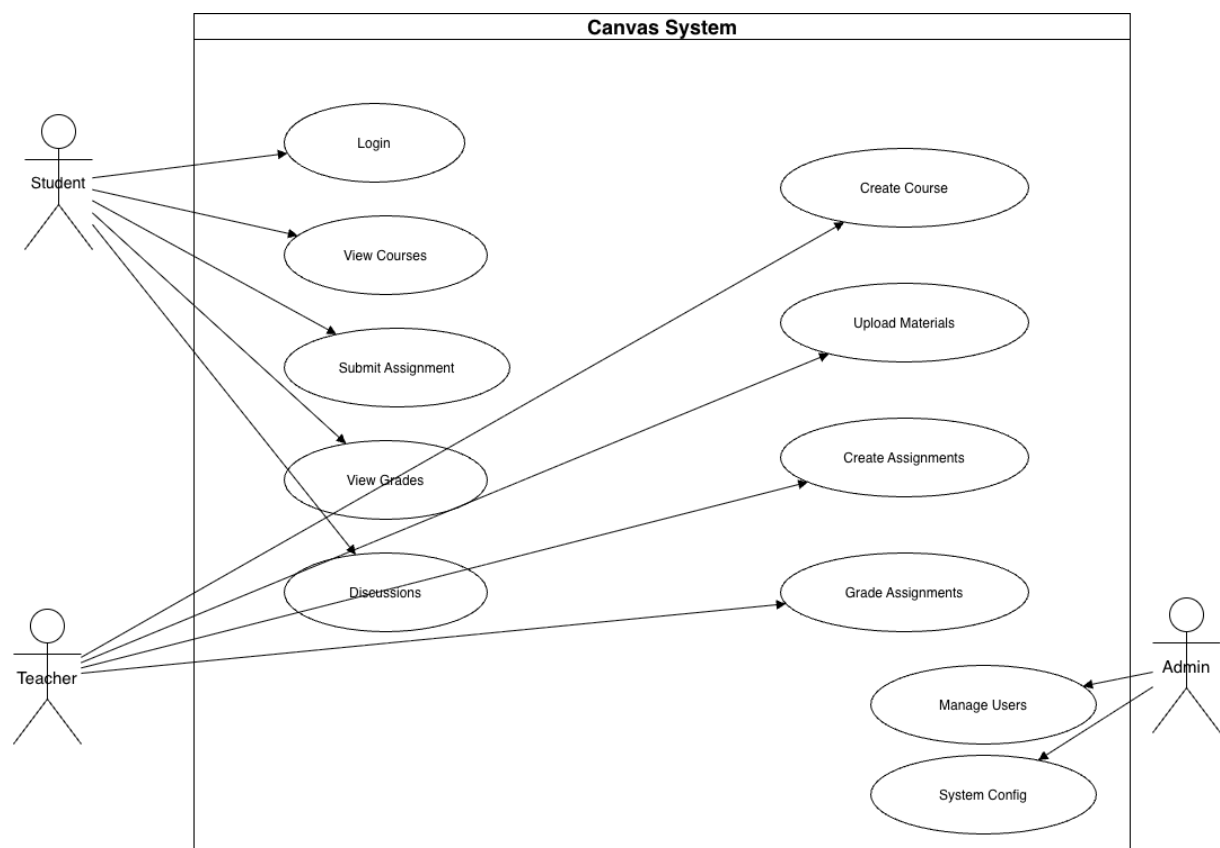
Question 6

Using the Agile Manifesto, Principle number 6 states “The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.”

- a) This principle means that the best way to share information is through direct, face-to-face communication, where team members can quickly ask questions and avoid misunderstandings.
- b) This is very important in helping the team to understand and to bounce ideas off of each other and also receive clear instructions from the systems architect or whoever did the requirements gathering.
- c) The Nearest principle to number six would be number 4 which states “Business people and developers must work together daily throughout the project.”. While working closely together, the developers will be able to flesh out any misunderstanding.

Question 7

Use Case Diagram - Using of Canvas



Question 8

Wireframe – Instant Messaging App for Desktop

The wireframe below shows three screens, **The main screen with login** (Figure 1), A **registration screen** (Figure 2) and **the chat screen**(Figure 3). The wireframe was done using the Pencil App for Mac.

The wireframe depicts a desktop application window titled "CHAT A BOX". Inside the window, there is a header box stating "CHAT A BOX is Hartnell Campbell's Instant Messaging app for Desktop". Below this, a central box prompts the user to "LOG IN TO START MESSAGING". Underneath the prompt are two input fields: one labeled "USERNAME" and another for a password, represented by "*****". A teal "LOGIN" button is positioned below the password field. At the bottom of the login section, another box says "NOT YET REGISTERED - GO AHEAD!!", with a light blue "REGISTER NOW" button centered below it.

Figure 1

Question 8 (contd)

CHAT A BOX

REGISTRATION

Signup now to enjoy free messaging with CHAT A BOX

ENTER FIRSTNAME

ENTER LASTNAME

ENTER EMAIL

ENTER PASSWORD

CONFIRM PASSWORD

☐ I AM OVER 18

☐ I AM HUMAN

REGISTER NOW

Figure 2

Question 8 (contd)

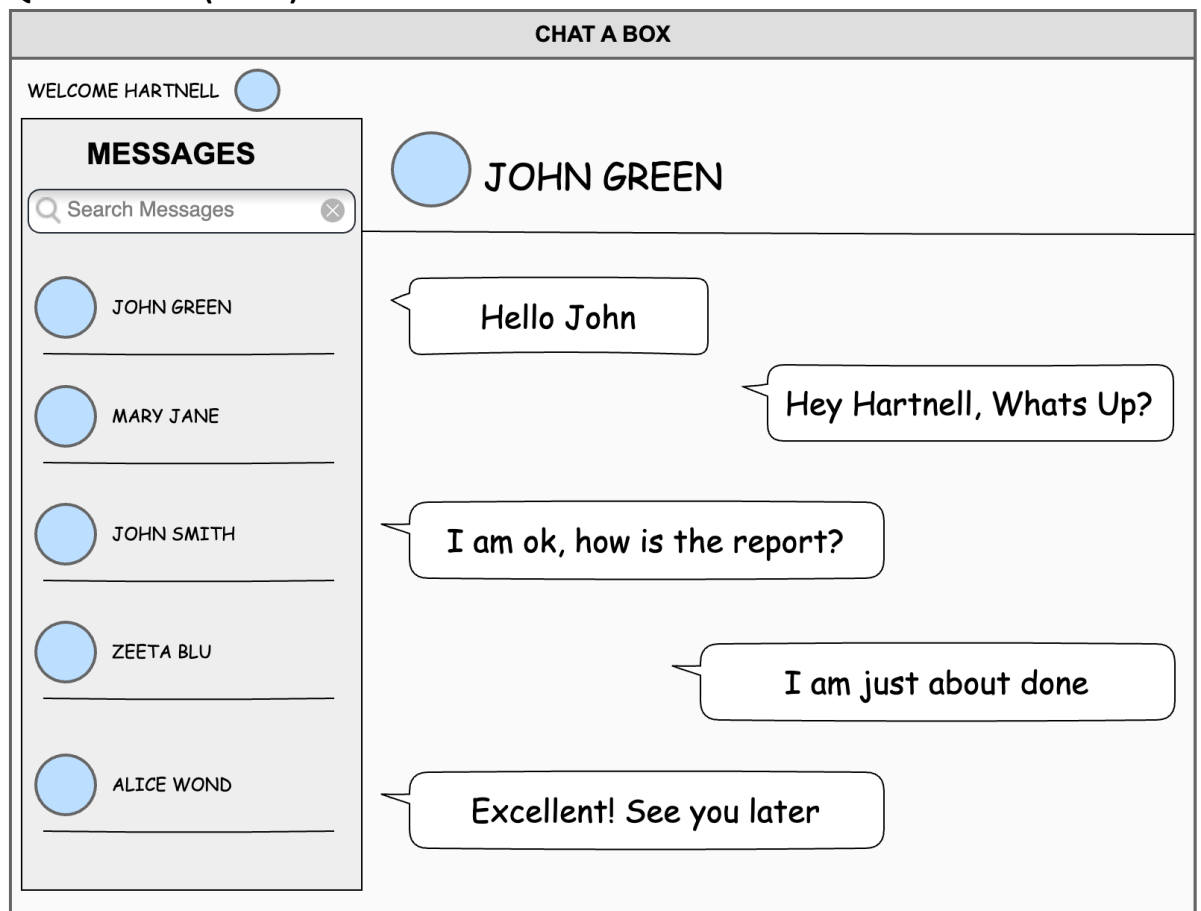
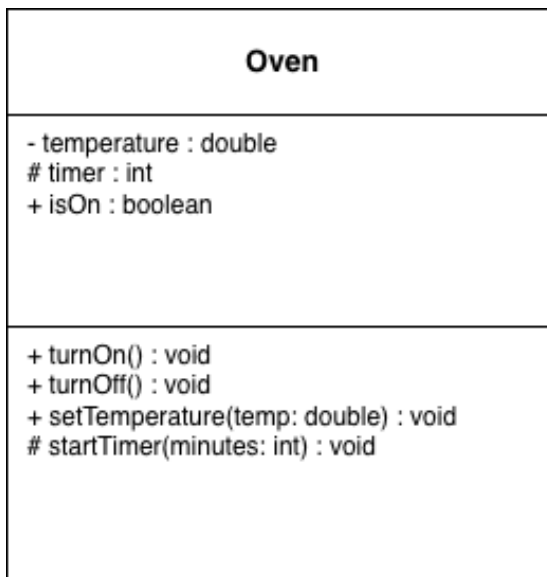


Figure 3

Question 9 – Oven Class Diagram



Question 10

CodeIgniter is a PHP Framework that uses the Model, View, Controller (MVC) architectural design.

The MVC design separates the software into three connected components to improve organization, scalability and maintainability:

- **The Model** interacts with the database to do work such as perform CRUD operations.
- **The View** is responsible for displaying the data to the user
- **The Controller** acts as the go-between the two Model and the View

Using the MVC architecture means there is cleaner, more organized code, its easier to make updates to areas without causing harm to the application, teams can work independently of each other on the same application and its much easier to extend the application(scalability).

Question 11

This assignment was done in Visual Studio Code

```
package default;  
  
/**  
 * The MyClass class provides basic is doing some basic  
 arithmetic operations  
 * and simple string manipulation methods.  
 *  
 * This class demonstrates common operations such as  
 addition, subtraction,
```



```

    * multiplication, division, modulus, and also showing
    greetings .
    */
    public class MyClass {

        /**
         * Adds two integer values.
         *
         * @param a the first integer
         * @param b the second integer
         * @return the sum of a and b
         */
        public int add(int a, int b){
            return a + b;
        }

        /**
         * Concatenates two strings.
         *
         * @param a the first string
         * @param b the second string
         * @return the result of a plus b
         */
        public String add(String a, String b){
            return a + b;
        }

        /**
         * Subtracts one integer from another.
         *
         * @param a the number to subtract from
         * @param b the number to subtract
         * @return the result of a minus b
         */
        public int sub(int a, int b){
            return a - b;
        }

        /**
         * Multiplies two integer values.
         *

```

```

    * @param a the first integer
    * @param b the second integer
    * @return the product of a and b
    */
    public int mult(int a, int b){
        return a * b;
    }

    /**
     * This function divides one integer by another.
     * Note: This method performs integer division.
     * If b is zero, it will throw an Exception.
     *
     * @param a the dividend
     * @param b the divisor
     * @return the result of a divided by b
     */
    public int div(int a, int b){
        return a / b;
    }

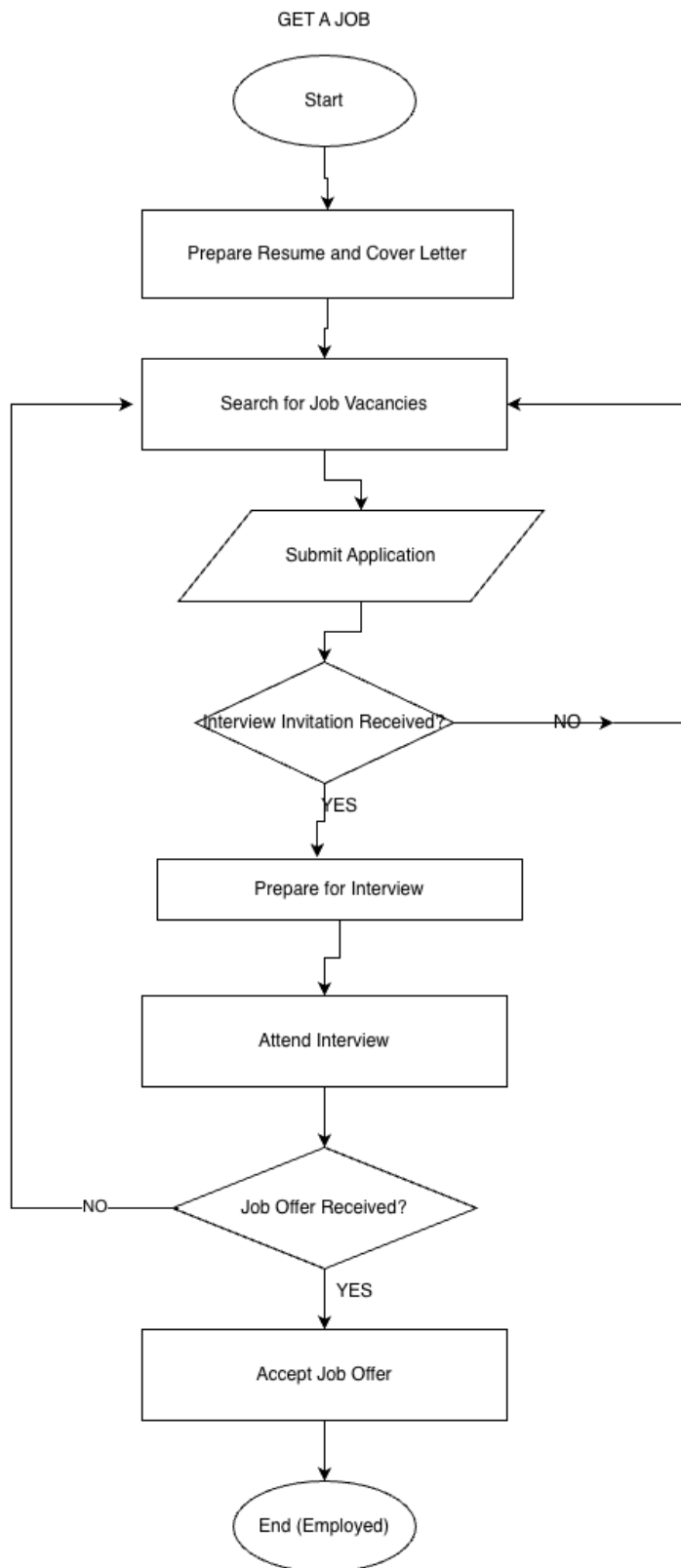
    /**
     * Calculates the modulus (remainder) of two integers.
     *
     * @param a the dividend
     * @param b the divisor
     * @return the remainder when a is divided by b
     */
    public int mod(int a, int b){
        return a % b;
    }

    /**
     * Generates a greeting message.
     *
     * @param name the name of the person to greet
     * @return a greeting string in the format "Hello
[name]"
     */
    public String sayHello(String name){
        return "Hello " + name;
    }

```

Question 12

Flowchart for getting a job



Question 13

The link to my repository is <https://github.com/MadPanta/UofH>